Yunbei Zhang

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Research Statement

I develop **multimodal LLMs and agents** with strong **image and video reasoning** through **RL post-training**, **agentic evaluation**, and inference-time adaptation. My work improves visual grounding, long-horizon decision-making, and reliability under distribution shift in open-world and safety-critical environments.

Education

- **Tulane University** — *Aug 2022 – Aug 2027 (Expected)*
  Ph.D. in Computer Science — New Orleans, LA, USA
- **Osaka University** — *Apr 2018 – Mar 2022*
  B.S. in Mathematics (Award of Excellence and multiple merit-based scholarships) — Osaka, Japan
- **Peking University** — *Jul 2013 – May 2015*
  Undergraduate Studies. Incomplete (Withdrew after two years to pursue alternative academic opportunities) — Beijing, China

Research Experience

- **Waymo** — *Jun 2026 – Aug 2026*
  Research Scientist Intern – Generative Planning & Post-Training (Mentor: Chi-Pang Lam and Hengbo Ma) — Mountain View, CA, USA
  - Developed **RL-based post-training for flow-matching trajectory generation models** in autonomous driving, designing collision-aware and scenario-conditioned rewards with trajectory-aware policy optimization to improve **long-horizon planning and rare-event robustness** in safety-critical long-tail scenarios, resulting in an **ICRA submission**.
- **Oak Ridge National Laboratory** — *Jan 2026 – May 2026*
  Graduate Research Intern – Foundation Models & Uncertainty Quantification (Mentor: Xiao Wang) — Oak Ridge, TN, USA
  - Developed **uncertainty quantification and efficient adaptation methods for large foundation models**, including weather forecasting models, using HPC to study predictive calibration, residual routing, and reliability under distribution shift. This work resulted in **two ECCV 2026 papers and a NeurIPS 2026 paper**.
- **Amazon** — *Sep 2025 – Dec 2025*
  Applied Scientist Intern – Multimodal LLM Post-Training & Reasoning (Mentor: Chandan K. Reddy) — Seattle, WA, USA
  - Designed **RL-based post-training and agentic evaluation for frontier MLLMs** and built automated red-teaming pipelines with planning agents to surface failures in vision-dependent reasoning, improving scalable evaluation of multimodal reasoning, alignment, and robustness (**ICLRW 2026 Best Short Paper Award**).
- **KLA Corporation** — *May 2025 – Aug 2025*
  Machine Learning Research Intern – Efficient Vision Foundation Models (Mentor: Rajkumar Theagarajan) — Milpitas, CA, USA
  - **Distilled a large vision foundation model into a compact YOLO detector**, designing the distillation and optimization pipeline to improve **detection accuracy and inference efficiency**. Achieved an **18% improvement in detection mAP**, with stronger gains on **out-of-distribution data** while retaining efficient inference.

Selected Publications (* equal contribution, full list on website)

1. **Y. Zhang**, C. Cai, F. Liu, J. Hamm.
   Prime Once, then Reprogram Locally: An Efficient Alternative to Black-Box Service Model Adaptation.
   CVPR, 2026. [paper] [code] **[Highlight]** (3.6% selection rate)
2. **Y. Zhang**, A. Mehra, S. Niu, J. Hamm.
   DPCore: Dynamic Prompt Coreset for Continual Test-Time Adaptation.
   ICML, 2025. [paper] [code]
3. **Y. Zhang**, S. Niu, C. Cai, F. Liu, J. Hamm.
   Adapting in the Dark: Efficient and Stable Test-Time Adaptation for Black-Box Models.
   NeurIPS, 2026. ICLR 2026 Workshop on Test-Time Updates (TTU). [project] [code] **[Best Paper Award]** (1 of 186 submissions)
4. **Y. Zhang**, J. Wang, X. Xiao, J. Hamm, X. Wang.
   UtoMe: Observation-Uncertainty-Guided Token Merging for Weather Foundation Models.
   NeurIPS, 2026. [paper]
5. G. Chen, S. Niu, G. Li, **Y. Zhang**, S. Shan, C. Miao, J. Yang.
   EVA-0: Test-Time Model Evolution with Only Two Forward Passes per Sample.
   NeurIPS, 2026. [paper]
6. **Y. Zhang**, Y. Ge, W. Xu, Y. Xu, J. Hamm, C. K. Reddy.
   Visual Exclusivity Attacks: Automatic Multimodal Red Teaming via Agentic Planning.
   ICLR 2026 Workshop on Agents in the Wild (AIWILD) [project] [code] **[Best Short Paper Award]** (1 of 237 submissions)
7. **Y. Zhang***, Z. Jin*, Y. Liu, J. Wang, X. Zhang, et al.
   LIBERO-MAX: Do Robot Policies Adapt When the World Changes?
   NeurIPS 2026 Robotics World Modeling Workshop. [project] [paper] **[Oral]**
8. M. Liu, **Y. Zhang**, S. Liu, L. Wang, W. Zhang.
   Wan-R1: Verifiable-Reinforcement Learning for Video Reasoning.
   ECCV, 2026. [paper]

9. X. Hu, H. Ni, **Y. Zhang**, J. Hamm, Z. Li, Z. Ding.
Seeing Clearly, Reasoning Confidently: Plug-and-Play Remedies for Vision Language Model Blindness.
CVPR, 2026. [paper]

10. J. Wang, **Y. Zhang**, L. Zhao, X. Xiao, J. Hamm, X. Wang.
Less Tokens, Better Forecasts: Sparse Residual Routing for Efficient Weather Prediction.
ECCV, 2026. [paper]

11. X. Xiao, C. Liu, C.-T. Liao, **Y. Zhang**, Q. Lan, Y. Wei, L. Zhao, et al.
Staying VIGILant: Mitigating Visual Laziness via Counterfactual Visual Alignment in MLLMs.
ECCV, 2026. [paper]

12. X. Xiao, X. Li, **Y. Zhang**, C. Han, T. Liu, T. Wang, R. Jiang, J. Hamm, et al.
Layer-Specific Prompt Fusion Discovery via Differentiable Search in Vision Foundation Models.
ECCV, 2026. [paper]

13. Y. Li, X. Xiao, **Y. Zhang**, L. Zhao, Y. Li, A. Zhao, T. Wang, H. Xu, Y. Tian.
Prompted Information Bottlenecks: Rethinking Layer-Wise Information Allocation in Frozen Vision Transformers.
ACM MM, 2026. [paper]

14. X. Jiang, L. Zhao, X. Xiao, **Y. Zhang**, J. Wang, C. Ma, H. Li, Y. Wang, Y. Gong, O. Camps.
Dynamic Hub-and-Spoke Memory for Streaming Video Understanding.
EMNLP Findings, 2026. [paper]

15. **Y Zhang***, Y Zhang*, M. Li, J. Wang, Z. Jin, S. Liu, D. Zhao.
Simple Agentic Memory for Generalist Robot Policies.
Preprint, 2026. [project] [paper]

16. Z. Jin*, **Y. Zhang***, Y. Liu, M. Liu, B. Chen, W. Ye, S. Liu, M. Pavone.
DeepJEPA: Scaling World Models from Within.
Preprint, 2026. [paper] [project]

17. **Y. Zhang**, J. Wang, Y. Ge, W. Xu, J. Hamm, C. K. Reddy.
Stop Comparing LLM Agents without Disclosing the Harness.
Preprint, 2026. [project] [code]

18. **Y. Zhang**, S. Lyu, J. Wang, Y. Ge, J. Guo, J. Hamm, C. K. Reddy.
Deny Without Disabling: Authorization-Paired Evaluation and Control for Multi-Agent Systems.
NeurIPS 2026 Workshop on Agents in the Wild (AIWILD). [paper]

19. X. Xiao*, C. Ma*, **Y. Zhang***, C. Liu, Z. Wang, Y. Li, L. Zhao, et al.
Not All Directions Matter: Towards Structured and Task-Aware Low-Rank Model Adaptation.
ACL (Main), 2026. [paper]

20. J. Wang*, **Y. Zhang***, Z. Ding, J. Hamm.
Doctor Approved: Generating Medically Accurate Skin Disease Images through AI-Expert Feedback.
NeurIPS, 2025. [paper] [code]

21. X. Xiao*, **Y. Zhang***, X. Li, T. Wang, X. Wang, Y. Wei, J. Hamm, M. Xu.
Visual Instance-Aware Prompt Tuning.
ACM MM, 2025. [paper]

22. J. Wang, X. Hu, **Y. Zhang**, D. Almamy, V. Bamba, et al.
eSkinHealth: A Multimodal Dataset for Neglected Tropical Skin Diseases.
ACM MM Datasets Track, 2025. [paper]

23. Y. Chung, **Y. Zhang**, N. Marrouche, J. Hamm.
SoK: Can Synthetic Images Replace Real Data? A Survey of Utility and Privacy of Synthetic Image Generation.
USENIX Security, 2025. [paper]

24. A. Mehra, **Y. Zhang**, J. Hamm.
Understanding the Transferability of Representations via Task-Relatedness.
NeurIPS, 2024. [paper]

Selected Honors & Grants

Final-round Candidate, Anthropic Fellows (2026)
Tinker Research Grant, Thinking Machines Lab (2026)
Lambda Research Grant, Lambda (2025)
Outstanding TA Award, Tulane University (2024)
Monbukagakusho Honors Scholarship, MEXT (2018, 2021)
Ichikawa International Scholarship (2021)
School of Science Honor Scholarship, Osaka (2018–2021)
Award of Excellence, Osaka University (2019)

Technical Skills & Service

- **Programming & ML Frameworks:** **Python**, C++, Bash, LaTeX, NumPy, **PyTorch**, **JAX**, Hugging Face Transformers, TRL, PEFT, verl, DeepSpeed, vLLM, Weights & Biases.
- **Training Systems:** Multi-node training and evaluation of foundation models on **ORNL Frontier (AMD MI250X)** and **NVIDIA GPU clusters**, scaling workloads to **hundreds of compute nodes**; Slurm, AWS, and large-scale experiment management.
- **Reviewer:** NeurIPS 2025–2026 (**2025 Top Reviewer**), ICLR 2025–2026, ICML 2026 (**Gold Reviewer**), CVPR 2026, ECCV 2026, ACM MM 2025–2026, and TMLR.
- **Languages:** English (fluent), Japanese (fluent), Chinese (native).